

MEDINA PRINCE ABDULAZIZ INTL, Saudi Arabia

WMO: 404300

Lat: 24.553N

Lon: 39.705E

Elev: 2151

StdP: 13.59

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	49.9	52.3	1.1	6.3	88.3	6.4	8.3	84.3	22.1	71.1	19.8	67.9	5.1	50	0.463

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.5	113.4	65.9	111.8	65.3	110.8	65.0	72.2	99.0	70.1	99.9	68.6	99.8	9.1	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.0	96.7	82.5	60.9	86.5	79.8	58.9	80.4	78.8	37.3	102.5	35.4	101.4	34.1	102.3	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.8	18.3	16.2	DB	43.8	116.2	3.5	1.4	41.3	117.2	39.2	118.0	37.2	118.7	34.6	119.7	(4)
(5)			WB	33.7	75.0	3.0	2.3	31.5	76.7	29.8	78.0	28.1	79.3	25.9	80.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	83.9	65.5	69.7	75.7	83.7	91.2	96.9	97.7	99.3	95.8	86.8	75.7	68.2	(6)	
	DBStd	12.80	5.34	5.55	5.61	4.89	4.28	3.14	3.93	3.47	3.40	3.90	5.05	4.84	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	107	61	17	2	0	0	0	0	0	0	0	1	26	(9)	
	CDD50	12381	481	552	798	1010	1276	1407	1478	1529	1373	1141	772	564	(10)	
	CDD65	7013	76	150	335	560	811	957	1013	1064	923	676	323	125	(11)	
	CDH74	112766	564	1372	3768	7788	13518	17447	18284	19267	16343	10198	3315	902	(12)	
(13)	CDH80	77637	93	405	1659	4503	9281	13137	13825	14807	12053	6342	1376	156	(13)	
(14)	Wind	WSAvg	7.0	6.4	7.2	7.6	7.5	7.2	7.4	8.0	7.7	6.6	5.8	6.4	6.3	(14)
(15) Precipitation	PrecAvg	2.4	0.3	0.1	0.3	0.3	0.2	0.0	0.0	0.2	0.0	0.2	0.6	0.3	(15)	
	PrecMax	5.0	1.8	1.3	1.6	1.7	1.6	0.2	0.7	1.4	0.3	1.7	2.3	1.6	(16)	
	PrecMin	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
	PrecStd	1.2	0.4	0.3	0.4	0.4	0.4	0.0	0.2	0.3	0.0	0.4	0.7	0.4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.0	91.4	98.5	104.2	109.7	113.3	115.9	115.2	112.7	104.3	95.2	86.3	(19)	
		MCWB	59.3	60.6	60.1	62.6	64.4	64.7	66.8	67.0	65.5	63.2	61.8	59.9	(20)	
	2%	DB	82.5	87.8	94.6	100.8	107.7	111.5	113.4	113.4	110.5	102.4	92.8	84.0	(21)	
		MCWB	58.6	60.0	60.4	62.0	63.6	64.2	65.8	66.4	64.7	62.4	61.0	60.6	(22)	
	5%	DB	80.2	84.6	91.4	98.7	105.8	109.8	111.4	112.6	109.1	100.7	89.9	81.7	(23)	
		MCWB	58.1	58.9	59.7	61.6	63.1	63.8	65.2	66.1	64.3	62.0	60.9	59.2	(24)	
	10%	DB	77.1	82.2	88.2	96.4	103.8	109.0	109.3	110.9	107.3	98.8	87.8	79.0	(25)	
		MCWB	56.9	58.1	58.6	61.3	62.9	63.6	64.8	65.7	63.9	61.5	60.5	58.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.5	66.1	68.0	69.2	69.9	71.0	73.4	76.2	71.3	69.1	67.7	66.4	(27)	
		MCDB	76.1	82.5	86.0	89.2	95.0	105.5	100.2	97.6	95.7	90.6	82.8	77.6	(28)	
	2%	WB	63.5	64.3	65.2	67.1	68.0	68.6	70.2	73.3	68.8	67.1	66.1	64.4	(29)	
		MCDB	76.1	80.5	84.1	88.7	95.4	103.1	101.7	98.1	99.7	89.8	81.4	77.6	(30)	
	5%	WB	61.6	62.6	63.2	65.4	66.8	67.1	68.3	71.1	67.3	65.5	64.6	62.7	(31)	
		MCDB	75.1	79.2	83.6	88.6	95.1	102.8	104.0	100.2	100.7	90.3	80.8	77.3	(32)	
	10%	WB	59.6	60.8	61.4	63.8	65.4	65.3	67.0	69.1	66.0	64.2	63.2	60.8	(33)	
		MCDB	73.2	77.2	82.9	88.9	95.1	103.2	104.2	102.6	100.8	91.5	81.2	75.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	20.7	21.8	22.9	23.6	23.6	23.9	22.4	22.5	23.4	23.6	21.1	20.4	(35)	
		MCDBR	23.5	24.1	25.2	25.4	25.0	24.5	23.2	23.3	23.4	24.4	23.6	22.2	(36)	
		MCWBR	12.0	11.6	10.7	9.5	8.7	8.0	7.7	8.0	8.0	9.1	9.7	10.7	(37)	
	5% WB	MCDBR	19.4	20.7	21.3	22.7	22.8	24.0	22.7	22.9	22.9	22.2	19.7	19.2	(38)	
		MCWBR	11.7	11.8	11.3	9.7	8.5	9.1	8.8	9.3	8.4	8.9	8.9	10.1	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.369	0.420	0.463	0.517	0.539	0.525	0.539	0.593	0.531	0.482	0.430	0.391	(40)	
	taud		2.302	2.114	1.977	1.874	1.828	1.861	1.862	1.743	1.855	1.970	2.114	2.221	(41)	
	Ebn at Noon		280	272	266	254	247	248	245	232	243	249	255	266	(42)	
	Edh at Noon		37	48	57	65	68	65	65	73	64	54	44	39	(43)	
(44) All-Sky Solar Radiation	RadAvg		1427	1706	2045	2240	2374	2539	2442	2249	2083	1811	1484	1336	(44)	
	RadStd		80	71	83	81	109	50	66	75	74	56	66	43	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.76	N/A	-3.42	N/A	N/A	N/A	N/A	N/A	+241	+238
(47) Regional (1 neighbor)	+0.83	N/A	-1.55	N/A	N/A	N/A	N/A	N/A	+281	+275

Nomenclature: See separate page